On a Sufficient Condition for Observability of Nonlinear Switched Systems and Observer Design

Hyungbo Shim and Aneel Tanwani

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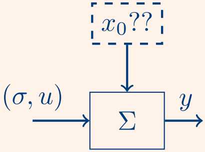
Switched Systems & Observability

Switched Jump Systems with Switching Signal $\sigma(t)$

$$\begin{aligned} \dot{x}(t) &= f_{\sigma(t)}(x(t)) + g_{\sigma(t)}(x(t))u(t) \\ x(t_q) &= p_{\sigma(t_q^-)}(x(t_q^-)) \\ y(t) &= h_{\sigma(t)}(x(t)) \end{aligned}$$



- Switching signal $\sigma : \mathbb{R} \mapsto \mathbb{N}$; piecewise constant, right-continuous, ever increasing
- Switching times $\{t_q\}$, $q \in \mathbb{N}$

Definition (Large-time observable on $\mathcal{X} \subset \mathbb{R}^n$)

$\exists T > t_0$ and $u_{[t_0,T]}$ s.t. $x(T)$ is determined uniquely from $y_{[t_0,T]}$, $u_{[t_0,T]}$, and $\sigma_{[t_0,T]}$ as long as $x_{[t_0,T]} \subseteq \mathcal{X}$.

- Small-time observability: if, in addition, $T > t_0$ is arbitrary.
- Add 'uniform' when observability is uniform w.r.t. input $u(t)$.

Overview

This talk is about

- Design of asymptotic observer for large-time observable system
  - Emphasis on the case when each mode is not observable (i.e., not small-time observable)
  - Development of a sufficient condition for large-time observability
  - Observer design strategy based on the sufficient condition

Related references:

- [Herman '77, Nijmeijer '90, Isidori '95]: local, instantaneous observability for non-switched nonlinear systems
- [Gauthier '92 & '94]: uniform observability w.r.t. inputs
- [Hespanha '05]: large-time observability of nonlinear systems
- [Vidal '03, Collins '04, Babaali '05]: recover discrete and continuous state simultaneously, use of derivatives of output
- [Xie '03, Sun '05]: large-time observability of switched linear systems
- [Balluchi '03, Tanwani '11]: large-time observability of switched linear systems and observer design

Review of Linear Case in [Tanwani/S/Liberzon; HSCC'11]

Switched linear sys. w/ jump:

$$\dot{x}(t) = A_q x(t)$$
$$x(t_q) = E_q x(t_q^-)$$
$$y(t) = C_q x(t)$$

Kalman decomposition at each mode:

$$\dot{\xi}_q' = F_q' \xi_q' + F_q'' \xi_q$$
$$\dot{\xi}_q = F_q \xi_q$$
$$y = H_q \xi_q$$
$$\begin{bmatrix} \xi_q'(t_q) \\ \xi_q(t_q) \end{bmatrix} = R_q \begin{bmatrix} \xi_{q-1}'(t_q^-) \\ \xi_{q-1}(t_q^-) \end{bmatrix}$$

The coordinate change yields:

$$\begin{aligned}
x(t_1^-) &= M_1 \xi_1(t_1^-) + N_1 \xi_1'(t_1^-) \\
x(t_2^-) &= M_2 \xi_2(t_2^-) + N_2 \xi_2'(t_2^-) \\
&\vdots \\
x(t_m^-) &= M_m \xi_m(t_m^-) + N_m \xi_m'(t_m^-) \\
&= \Psi_{m-1}^m \big(M_{m-1} \xi_{m-1}(t_{m-1}^-) \\
&\qquad + N_{m-1} \xi_{m-1}'(t_{m-1}^-) \big) \\
&= \cdots \\
&= \Psi_1^m \big(M_1 \xi_1(t_1^-) + N_1 \xi_1'(t_1^-) \big)
\end{aligned}$$

where

$$\Psi_i^j := e^{A_j(t_j - t_{j-1})} E_{j-1} \cdots e^{A_{i+1}(t_{i+1} - t_i)} E_i$$

Pick the matrices Θ_j s.t.

$$\text{span}\{\Theta_j\} = \text{span}\{\Psi_j^m N_j\}^\perp.$$

Then,

$$\begin{bmatrix} \Theta_m \\ \vdots \\ \Theta_2 \\ \Theta_1 \end{bmatrix} x(t_m^-) = \begin{bmatrix} \Theta_m \Psi_m^m M_m \xi_m(t_m^-) + \Theta_m \Psi_m^m N_m \xi_m'(t_m^-) \\ \vdots \\ \Theta_2 \Psi_2^m M_2 \xi_2(t_2^-) + \Theta_2 \Psi_2^m N_2 \xi_2'(t_2^-) \\ \Theta_1 \Psi_1^m M_1 \xi_1(t_1^-) + \Theta_1 \Psi_1^m N_1 \xi_1'(t_1^-) \end{bmatrix}.$$

It is shown in [HSCC'11] that

Left invertibility of $\begin{bmatrix} \Theta_m \\ \vdots \\ \Theta_2 \\ \Theta_1 \end{bmatrix} \iff$ Switched system is large-time observable.

This approach applies to linear systems only.

Example: Accumulating partial info up to some time t_2

$$\dot{x} = f_1(x) = \begin{pmatrix} x_3 \\ x_1^2 - x_3^2 + 2x_1 \\ x_1 + 1 \end{pmatrix}$$
$$y = h_1(x) = x_2$$
$$x^+ = p_1(x) = 0.05 \begin{pmatrix} x_1 \\ 2x_2 \\ x_3 \end{pmatrix}$$

$$\dot{x} = f_2(x) = \begin{pmatrix} x_3 \\ (x_1^2 - x_3^2 + 2x_1)x_2 \\ x_1 + 1 \end{pmatrix}$$
$$y = h_2(x) = x_1^2 - x_3^2 + 2x_1$$
$$x^+ = p_2(x) = x$$

$$\dot{x} = f_3(x) = \begin{pmatrix} x_2^2 \\ -\frac{1}{2}x_2 \\ 0 \end{pmatrix}$$
$$y = h_3(x) = x_1 + x_2^2$$

$$\mathcal{X} = \{(x_1, x_2, x_3) : x_1 > 0, x_3 > 0\}$$

$$\ddot{y} = L_{f_1}^2 h_1 = 0$$
$$\dot{y} = L_{f_2} h_2 = 0$$
$$\dot{y} = L_{f_3} h_3 = 0$$

no one is observable

Timeline: (t_0)—(t_1^-) (t_1)—(t_2^-) (t_2)—

$$\begin{aligned} x_2(t_2) &= x_2(t_2^-) \\ &= \exp\left(\int_{t_1}^{t_2^-} y(s)ds\right) x_2(t_1) \\ &= \exp\left(\int_{t_1}^{t_2} y(s)ds\right) (0.1y(t_1^-)) \end{aligned}$$

$$x_1(t_2) = y(t_2) - x_2^2(t_2)$$

$$\begin{aligned} x_3(t_2) &= x_3(t_2^-) \\ &= \pm\sqrt{x_1^2(t_2^-) + 2x_1(t_2^-) - y(t_2^-)} \\ &= +\sqrt{x_1^2(t_2) + 2x_1(t_2) - y(t_2^-)} \end{aligned}$$

Example: Accumulating partial info up to some time t_2

$$\ddot{y} = L_{f_1}^2 h_1 = 0$$
$$\dot{y} = L_{f_2} h_2 = 0 \quad \text{no one is observable}$$
$$\dot{y} = L_{f_3} h_3 = 0$$

Timeline: (t_0)—(t_1^-) (t_1)—(t_2^-) (t_2)—

$$x_2(t_2) = x_2(t_2^-) = \exp\left(\int_{t_1}^{t_2^-} y(s)ds\right) x_2(t_1) = \exp\left(\int_{t_1}^{t_2} y(s)ds\right)(0.1y(t_1^-))$$

$$x_1(t_2) = y(t_2) - x_2^2(t_2)$$

$$x_3(t_2) = x_3(t_2^-) = \pm\sqrt{x_1^2(t_2^-) + 2x_1(t_2^-) - y(t_2^-)} = +\sqrt{x_1^2(t_2) + 2x_1(t_2) - y(t_2^-)}$$

Lessons:

- To recover $x(t_2)$, information obtained at each individual mode is collected by transporting through system dynamics.
- During the transportation through the continuous dynamics or the jump map, the obtained information is not corrupted by unobservable quantities.

"Switched Observable Canonical Structure"

Suppose that each mode of the system is transformed as:

(Mode 1): $[t_0, t_1)$

$$\dot{\xi}_1' = F_1'(\xi_1', \xi_1) + G_1'(\xi_1', \xi_1)u$$
$$\dot{\xi}_1 = F_1(\xi_1) + G_1(\xi_1)u$$
$$y = H_1(\xi_1)$$

(Mode 2): $[t_1, t_2)$

$$\dot{\xi}_2' = F_2'(\xi_2', z_2, \xi_2) + G_2'(\xi_2', z_2, \xi_2)u$$
$$\dot{z}_2 = F_2^*(z_2, \xi_2) + G_2^*(z_2, \xi_2)u$$
$$\dot{\xi}_2 = F_2(\xi_2) + G_2(\xi_2)u$$
$$y = H_2(\xi_2)$$
$$z_2(t_1) = R_2(\xi_1(t_1^-)) = R_2(\xi_1(t_1^-), \xi_2(t_1))$$

(Mode 3): $[t_2, t_3)$

$$\vdots$$
$$z_3(t_2) = R_3(\xi_2(t_2^-), z_2(t_2^-), \xi_3(t_2))$$
$$\vdots$$

(Mode m): $[t_{m-1}, t_m)$

there's no ξ_m'
and $\dim(z_m, \xi_m) = n$.

IF $\xi_q(t)$ is known for each $[t_{q-1}, t_q)$,

THEN $x(T)$, $t_{m-1} < T < t_m$, is determined by inverse transformation from $(z_m(T), \xi_m(T))$.

Sufficient Condition for Large-time Observability [local version]

Without state jumps ($\forall p_q(x) = x$):

- construct the observation space $d\mathcal{O}_q = \text{span}\{d\lambda_{q,i}(x) : 1 \leq i \leq k_q\}$, where $\lambda_{q,i}(x) \in \{h_q(x), L_{f_q}h_q(x), L_{g_q}h_q(x), L_{g_q}L_{f_q}h_q(x), \dots\}$
- let $\mathcal{W}_0 := \{0\}$ and $\mathcal{W}_q := \langle \mathcal{W}_{q-1} + d\mathcal{O}_q | f_q, g_q \rangle$

if $\exists$ m s.t. $\dim \mathcal{W}_m = n$, then Σ admits Switched Obs. Canonical Form.

With state jumps:

- introduce $d\mathcal{O}_q' := \text{span}\{d(\lambda_{q,i} \circ p_{q-1}) : 1 \leq i \leq k_q\}$ for each $q \geq 2$
- $\mathcal{W}_q := \langle \mathcal{W}_{q-1} + d\mathcal{O}_q | f_q, g_q \rangle$ such that

$$(p_q)_*(\ker \mathcal{W}_q \cap \ker d\mathcal{O}_{q+1}') \subset \ker \mathcal{W}_q$$

if $\exists$ m s.t. $\dim \mathcal{W}_m = n$, then Σ admits Switched Obs. Canonical Form.

Switched Obs. Canonical Form $\Rightarrow$ Large-time Observability

Example (continued)

Mode 1:

$\xi_{1,1} := \lambda_{1,1} = h_1(x)$

$\xi_{1,2} := \lambda_{1,2} = L_{f_1} h_1(x)$

$\mathcal{W}_1 = d\mathcal{O}_1 =$
row.span $\left\{ \begin{bmatrix} 0 & 1 & 0 \\ x_1 + 1 & 0 & -x_3 \end{bmatrix} \right\}$

Mode 2:

$\xi_{2,1} := \lambda_{2,1} = h_2(x) = L_{f_1} h_1(x)$

$\mathcal{W}_2 = \mathcal{W}_1$

Mode 3:

$\xi_{3,1} := \lambda_{3,1} = h_3(x)$

$\mathcal{W}_3 = \mathcal{W}_2 + d\mathcal{O}_3 = \mathbb{R}^n =$
row.span $\left\{ \begin{bmatrix} 0 & 1 & 0 \\ x_1 + 1 & 0 & -x_3 \\ 1 & 2x_2 & 0 \end{bmatrix} \right\}$

Dynamics over $[t_0, t_1)$:

$\dot{\xi}_{1,1} = \xi_{1,2}$

$\dot{\xi}_{1,2} = 0$

$y = \xi_{1,1}$

Dynamics over $[t_1, t_2)$:

$\dot{\xi}_{2,1} = 0$

$y = \xi_{2,1}$

$\dot{z}_{2,1} = z_{2,1}\xi_{2,1}$

$z_{2,1}(t_1) = 0.1\xi_{1,1}(t_1^-)$

Dynamics over $[t_2, t_3)$:

$\dot{\xi}_{3,1} = 0$

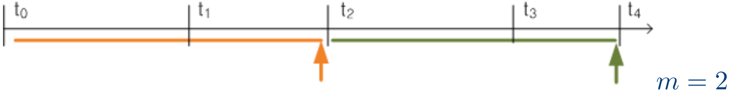
$y = \xi_{3,1}$

$\dot{z}_{3,1} = -\frac{1}{2} z_{3,1}$

$\dot{z}_{3,2} = 2(\xi_{3,1} - z_{3,1}^2 + 1) z_{3,1}^2$

$z_{3,1}(t_2) = z_{2,1}(t_2^-), \; z_{3,2}(t_2) = \xi_{2,1}(t_2^-)$

Observer Design



Assumption

- *Persistent switching and* $\exists D$ *s.t.* $t_q - t_{q-1} \le D$, $^\forall q \in \mathbb{N}$.
- $\exists m$ *s.t. the mode sequence* $1 \to 2 \to \cdots \to m$ *ensures large-time* uniform *observability, and the sequence repeats.*
- $^\forall t \ge t_0$, $x(t) \in \mathcal{X}$*: compact set, and* $|u(t)| \le M_u$.

Synchronous Observer (running parallel to the plant)

$$\dot{\hat{x}}(t) = \bar{f}_q(\hat{x}(t)) + \bar{g}_q(\hat{x}(t))u(t), \qquad t \in [t_{q-1}, t_q),$$

$$\hat{x}(t_q) = \begin{cases} \bar{p}_q(\hat{x}(t_q^-)), & (q \bmod m) \neq 0, \\ \bar{p}_q(x^\dagger(t_q^-)), & (q \bmod m) = 0, \end{cases}$$

- $x^\dagger(t_q^-)$ is computed at time t_q^- from the stored $y_{[t_{q-m}, t_q)}$ and $u_{[t_{q-m}, t_q)}$.
- $\bar{f}_q(x)$ is globally Lipschitz s.t. $\bar{f}_q(x) = f_q(x)$ on $\mathcal{X}$. Similar for others.

Computing $x^\dagger(t_m^-)$ from y and u on $[t_0, t_m)$

Plant:

(Mode 1): $[t_0, t_1)$

$$\dot{\xi}_1 = F_1(\xi_1) + G_1(\xi_1)u$$
$$y = H_1(\xi_1)$$

(Mode 2): $[t_1, t_2)$

$$\dot{\xi}_2 = F_2(\xi_2) + G_2(\xi_2)u$$
$$y = H_2(\xi_2)$$
$$\dot{z}_2 = F_2^*(z_2, \xi_2) + G_2^*(z_2, \xi_2)u$$
$$z_2(t_1) = R_2(\xi_1(t_1^-), \xi_2(t_1))$$
$$\vdots$$

Observing method requires that $\hat{\xi}_q(t) \approx \xi_q(t)$ for $[t_{q-1}, t_q)$.

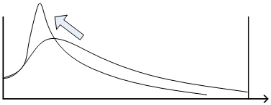
Observer:

(Mode 1): $\hat{\xi}_1 = \bar{\phi}_1(\hat{x}(t_0))$

$$\dot{\hat{\xi}}_1 = \bar{F}_1(\hat{\xi}_1) + \bar{G}_1(\hat{\xi}_1)u + L_1(\hat{\xi}_1, u, y) \cdot (y - \bar{H}_1(\hat{\xi}_1))$$

(Mode 2): $\hat{\xi}_2 = \bar{\phi}_2(\hat{x}(t_1))$

$$\dot{\hat{\xi}}_2 = \bar{F}_2(\hat{\xi}_2) + \bar{G}_2(\hat{\xi}_2)u + L_2(\hat{\xi}_2, u, y) \cdot (y - \bar{H}_2(\hat{\xi}_2))$$
$$\dot{\hat{z}}_2 = \bar{F}_2^*(\hat{z}_2, \hat{\xi}_2) + \bar{G}_2^*(\hat{z}_2, \hat{\xi}_2)u$$
$$\hat{z}_2(t_1) = \bar{R}_2(\hat{\xi}_1(t_1^-), \hat{\xi}_2(t_1))$$

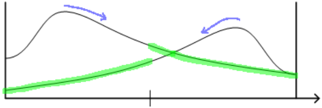
$\hat{\xi}_2(t) - \xi_2(t)$ 

Idea of 2-pass Back-and-forth Observer

Forward observer: with $\hat{\xi}_2^f = \bar{\phi}_2(\hat{x}(t_1))$, on $[t_1, t_2)$,

$$\dot{\hat{\xi}}_2^f = \bar{F}_2(\hat{\xi}_2^f) + \bar{G}_2(\hat{\xi}_2^f)u + L_2^f(\hat{\xi}_2^f, u, y) \cdot (y - \bar{H}_2(\hat{\xi}_2^f))$$

Estimation error:



Backward observer: with $\hat{\xi}_2^b(0) = \hat{\xi}_2^f(t_2^-)$,

for $s \in [0, t_2 - t_1)$,

$$\dot{\hat{\xi}}_2^b = -\bar{F}_2(\hat{\xi}_2^b) - \bar{G}_2(\hat{\xi}_2^b)u(t_2 - s) - L_2^b(\hat{\xi}_2^b, u, y) \cdot (y(t_2 - s) - \bar{H}_2(\hat{\xi}_2^b))$$

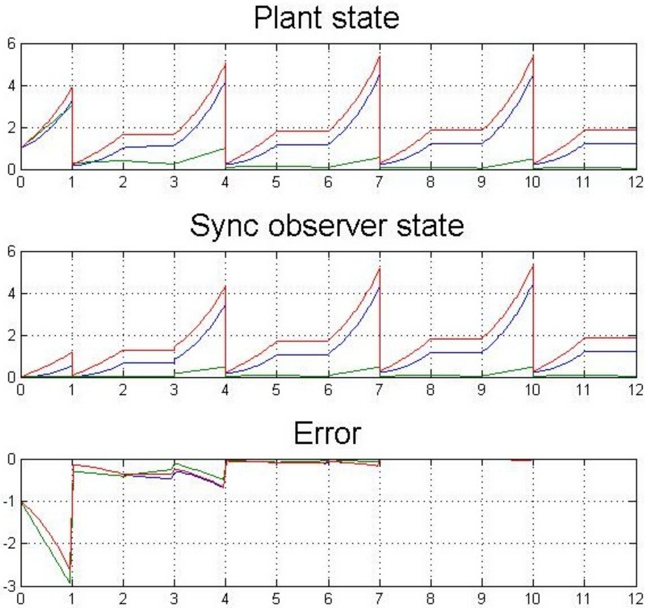
Convergence Result

At time t_m^-, set $x^\dagger(t_m^-) = \bar{\phi}_m^{-1}(\hat{\xi}_m(t_m^-), \hat{z}_m(t_m^-))$. Then,

$$\begin{aligned} |\hat{x}(t) - x(t)| &\leq \Gamma |\hat{x}(t_0) - x(t_0)|, && t_0 \leq t < t_m \\ |\hat{x}(t_m) - x(t_m)| &\leq \gamma |\hat{x}(t_0) - x(t_0)|, && 0 < \gamma < 1. \end{aligned}$$

Repeating the process, it follows that $\lim_{t\to\infty} |\hat{x}(t) - x(t)| = 0$.

Example: Simulation result



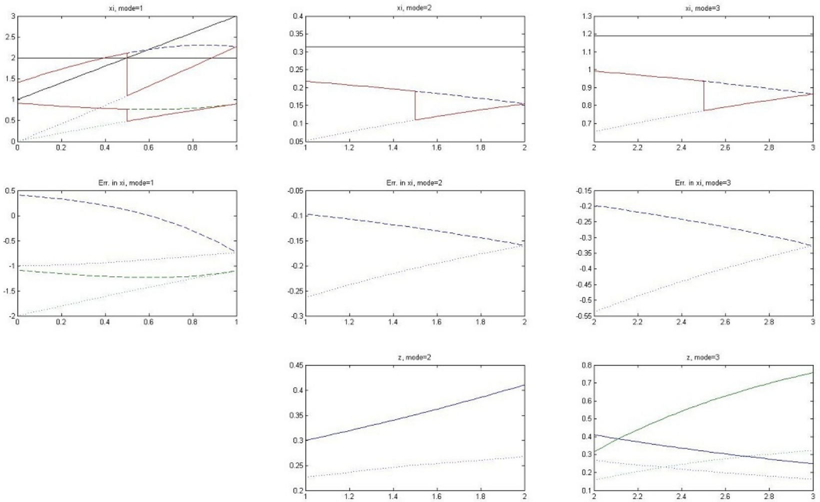
Conclusions & Future Work

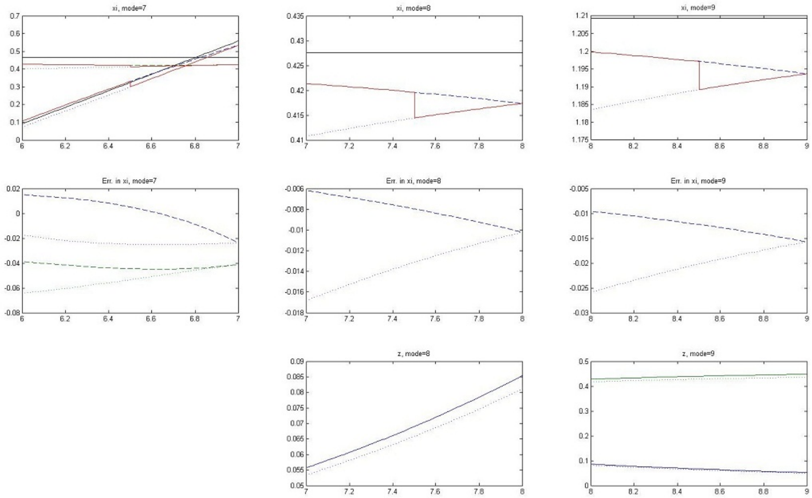
We have presented

- A sufficient condition for observability of switched nonlinear systems
- Asymptotic state estimation with an observer under persistent switching
- Novel idea of back-and-forth observer

Future works:

- Consideration of computation time
- Study the applications of current work


xi, mode=1
xi, mode=2
xi, mode=3
Err. in xi, mode=1
Err. in xi, mode=2
Err. in xi, mode=3
z, mode=2
z, mode=3


xi, mode=7
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
6
6.2
6.4
6.6
6.8
7
xi, mode=8
0.435
0.43
0.425
0.42
0.415
0.41
7
7.2
7.4
7.6
7.8
8
xi, mode=9
1.21
1.205
1.2
1.195
1.19
1.185
1.18
1.175
8
8.2
8.4
8.6
8.8
9
Err. in xi, mode=7
0.02
0
-0.02
-0.04
-0.06
-0.08
6
6.2
6.4
6.6
6.8
7
Err. in xi, mode=8
-0.006
-0.008
-0.01
-0.012
-0.014
-0.016
-0.018
7
7.2
7.4
7.6
7.8
8
Err. in xi, mode=9
-0.005
-0.01
-0.015
-0.02
-0.025
-0.03
8
8.2
8.4
8.6
8.8
9
z, mode=8
0.09
0.085
0.08
0.075
0.07
0.065
0.06
0.055
0.05
7
7.2
7.4
7.6
7.8
8
z, mode=9
0.5
0.4
0.3
0.2
0.1
0
8
8.2
8.4
8.6
8.8
9